

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Supplementary Examination June 2023 (2019 scheme)

Course Code: EET204**Course Name: ELECTROMAGNETIC THEORY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|---|-----|
| 1 | Explain the physical meaning of the divergence of a vector field. | (3) |
| 2 | Given the vector field $\vec{A} = x^2 \hat{z}$ where $\hat{z}$ denotes the unit vector along the z-axis in a right-handed Cartesian co-ordinate system. Compute $\vec{\nabla} \times \vec{A}$. | (3) |
| 3 | State Coulombs law and use it to represent the vector forces $\vec{F}_1$ and $\vec{F}_2$ acting on two point charges q_1 and q_2 respectively, separated by a distance r in free space. | (3) |
| 4 | Define electric field intensity and electric potential. | (3) |
| 5 | Explain the terms self-inductance and mutual inductance | (3) |
| 6 | State Amperes circuital law and express it in integral and vector forms | (3) |
| 7 | Define the term intrinsic impedance. | (3) |
| 8 | Define the term skin depth. | (3) |
| 9 | Explain impedance matching. | (3) |
| 10 | What are the various transmission line parameters? | (3) |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|------|
| 11 | a) State Stokes Theorem. | (4) |
| | b) Consider a vector field represented in a right-handed Cartesian coordinate system as $\vec{f} = xy \hat{x} + yz \hat{y}$ where $\hat{x}$ and $\hat{y}$ represent the unit vectors along the x and y directions respectively. Compute $\vec{\nabla} \cdot \vec{f}$ and verify the divergence theorem for the unit cubical volume shown in Fig. 1. | (10) |

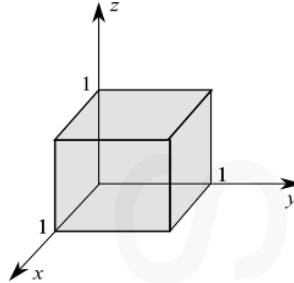


Fig. 1

- 12 a) Derive the matrix which transforms the vector $\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$ expressed in the Cartesian co-ordinates, to $\vec{A} = A_\rho \hat{\rho} + A_\phi \hat{\phi} + A_z \hat{z}$ as expressed in cylindrical co-ordinates. (6)
- b) Using spherical co-ordinates, show that the volume V of a sphere of radius R is given as $V = \frac{4}{3} \pi R^3$ (8)

Module -2

- 13 a) Explain Poisson's and Laplace's equations. (4)
- b) A circular disc of radius 'a' m is charged uniformly with a charge density of ' σ ' Coulombs/ m² lying in the xy -plane and centred at the origin of the coordinate system. Find the electric field at a point on the z -axis at a distance h above the plane (10)

- 14 a)

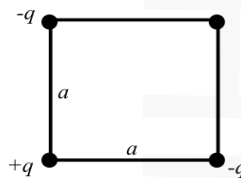


Fig. 2

Three charges are located at the corners of a square of side a , as shown in Fig.2. Calculate the work done in bringing another charge $+q$, from a far-away point and placing it at the fourth corner.

- b) Using Gauss's law, derive an expression for the electric field intensity due to an infinite line charge with charge density λ Coulombs per meter. (7)

Module -3

- 15 a) A circular loop of wire lying in the xy -plane and centred at the origin of the coordinate system is carrying a current I . Using Biot and Savart's law calculate the magnetic flux density at a point on the z -axis at a distance h above the plane of the loop. (7)

- b) Derive the boundary conditions that must be obeyed by static $\vec{H}$ and $\vec{B}$ fields at the interface of two different materials of permeability μ_1 and μ_2 . (7)
- 16 a) Explain the formulation of the magnetic vector potential $\vec{A}$. Why is $\vec{\nabla} \cdot \vec{A}$ chosen as equal to zero? (7)
- b) Using Biot-Savart's law, derive an expression for the magnetic flux density due to an infinitely long current carrying conductor in free space lying along the z-axis. Use the result to calculate the force per unit length between two infinitely long conductors carrying a current of I A each in the z-direction separated by a distance of l m. (7)

Module -4

- 17 a) State and prove Poynting's theorem. Explain the physical significance of the Poynting vector. (7)
- b) The electric field inside a non-magnetic medium is given as:

$$\vec{E} = 50 \cos(8x - 10^9 t) \hat{y} - 40 \sin(8x - 10^9 t) \hat{z} \text{ V/m} \quad (7)$$

Calculate the corresponding $\vec{H}$ field.

- 18 a) Starting from the Maxwell's equations, show that empty space can support the propagation of electromagnetic waves, and calculate the speed of these waves. (7)
- b) Derive the expression for the attenuation constant and phase constant for a uniform plane wave in a good conductor (7)

Module -5

- 19 a) What are the conditions required for a distortion less transmission line. Derive the phase velocity of such a line and explain its benefits. (6)
- b) Derive the expression for the input impedance and the voltage reflection coefficient for a lossless transmission line of characteristic impedance Z_0 terminated with load impedance Z_L . (8)
- 20 a) A lossless line operating at 100 MHz has a characteristic impedance $Z_0 = 70 \Omega$ and phase constant of 3 rad/m . Calculate the inductance per meter and capacitance per meter for the line. (6)
- b) Derive the equation representing the propagation of a voltage wave on a transmission line. (8)
